ENTERPRISE INFRASTRUCTURE PLAN | ABC Startup Solutions

ITEP 414 – System Administration and Maintenance

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11 August 2026

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Part 1: Company Profile

Company Name: ABC Startup Solutions

Nature of Business: Software development and IT consulting services for small and medium-sized businesses.

Company Vision:

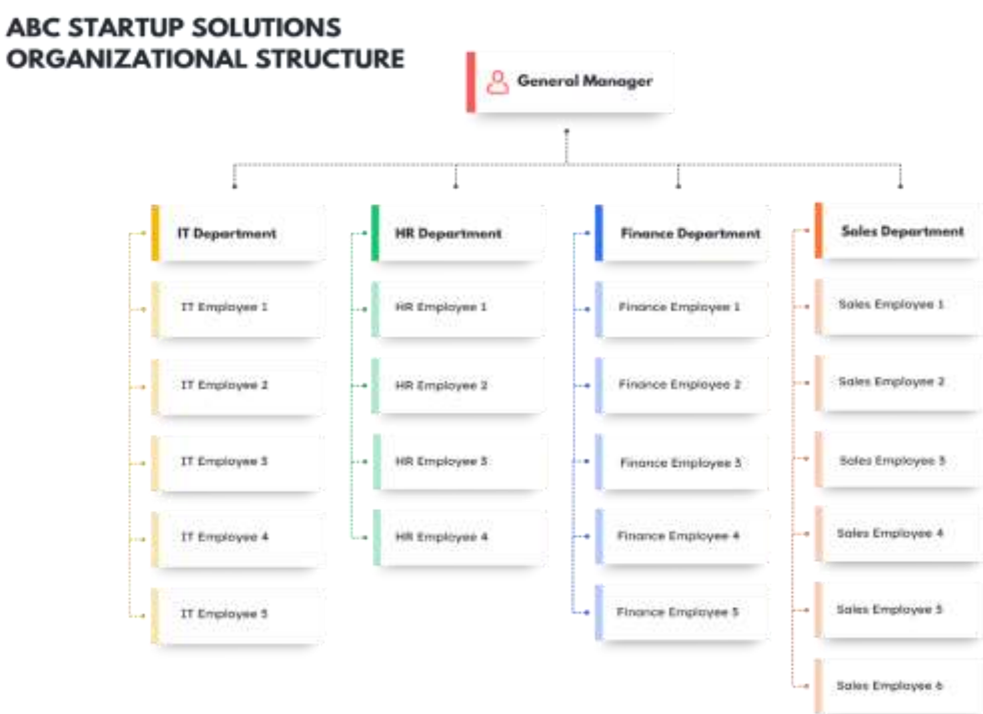
To become a trusted provider of innovative and secure software solutions that help businesses improve productivity, strengthen their digital capabilities, and achieve successful digital transformation.

Company Mission:

Our mission is to deliver reliable software products and IT services while maintaining secure and efficient IT infrastructure. We aim to support business growth through technology-driven solutions and promote continuous learning, innovation, and professional development among our employees.

Office Location (Fictional): 4th Floor, Meridian Tech Building, 8577 Makati Avenue, Poblacion, Makati City, Metro Manila, Philippines

Organizational Structure:



Employee Distribution:

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
TOTAL	**20**

Part 2: Enterprise Hardware Inventory

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer	12	HR, Finance, Sales	Daily Office
HW-002	Laptop	8	IT & Management	Mobility and remote support
HW-003	Rack Server	1	IT	Centralized file, application, and authentication services
HW-004	Router	1	IT	Internet connectivity and traffic routing
HW-005	Managed Switch (24-port)	1	IT	Connect all wired network devices
HW-006	Network Printer	2	Shared	Printing and document scanning
HW-007	UPS 1500VA	2	IT & Finance	Power protection for critical systems
HW-008	Wireless Access Point	3	Office-wide	Wireless network coverage
HW-009	NAS Storage	1	IT	Centralized file storage and backup
HW-0010	External Backup Drive (4TB)	2	IT	Offline backup and disaster recovery
HW-0011	24-inch Monitor	20	All Departments	Dual-display productivity and workstation use

Hardware Justification:

- 12 desktops for fixed office users and daily business operations.
- 8 laptops for IT staff, management, and mobile work requirements.
- 1 rack server for centralized files, applications, and user authentication.
- 1 router for internet connectivity, traffic routing, and basic network security.
- 1 managed switch for connecting wired devices and supporting future expansion.
- 2 network printers for shared printing and scanning across departments.

- 2 UPS units for protecting critical equipment from power interruptions and surges.
- 3 access points for stable wireless coverage across the office floor.
- 1 NAS storage for centralized file storage and backup management.
- 2 external 4TB drives for offline backups and disaster recovery.
- 20 monitors for dual-display workstations and improved employee productivity.

Part 3: Enterprise Software Inventory

Software	Version	License	Purpose
Windows 11 Pro	24H2	Commercial	Desktop operating system
Ubuntu Server	24.04 LTS	Open Source	Server operating system
Microsoft 365 Apps	Version 2606	Subscription	Productivity applications
Visual Studio Code	1.131	Free	Source code editor
Git	2.50.1	Open Source	Version control
GitHub Desktop	3.6.3	Free	GitHub repository management
Oracle VirtualBox	7.2.0	Open Source	Virtual machine virtualization
Google Chrome	151.0.7922.108/.109	Free	Web browser
Microsoft Defender Antivirus	Current platform version	Included	Endpoint security and antivirus
AnyDesk	9.7.12	Free/Commercial	Remote technical support
7-Zip	26.00	Open Source	File compression and extraction

Software Justification:

- **Windows 11 Pro** – Provides a secure and reliable operating system for employee workstations and supports business management features.
- **Ubuntu Server** – Provides a stable and cost-effective server operating system for hosting company services and applications.
- **Microsoft 365 Apps** – Supports daily productivity through Word, Excel, PowerPoint, Outlook, and other business applications.
- **Visual Studio Code** – Provides developers with a lightweight environment for writing, editing, and debugging source code.
- **Git** – Allows developers to track code changes, manage versions, and collaborate safely on software projects.

- **GitHub Desktop** – Simplifies Git and GitHub repository management for employees who prefer a graphical interface.
- **Oracle VirtualBox** – Allows IT staff to create virtual machines for testing software, operating systems, and configurations.
- **Google Chrome** – Provides employees with a reliable web browser for accessing websites, cloud applications, and online business systems.
- **Microsoft Defender Antivirus** – Protects company computers against malware, viruses, and other security threats.
- **AnyDesk** – Allows IT personnel to remotely access and troubleshoot employee computers when technical assistance is required.
- **7-Zip** – Provides file compression and extraction capabilities, helping employees manage and transfer large files efficiently.

Part 4: Enterprise Network Inventory

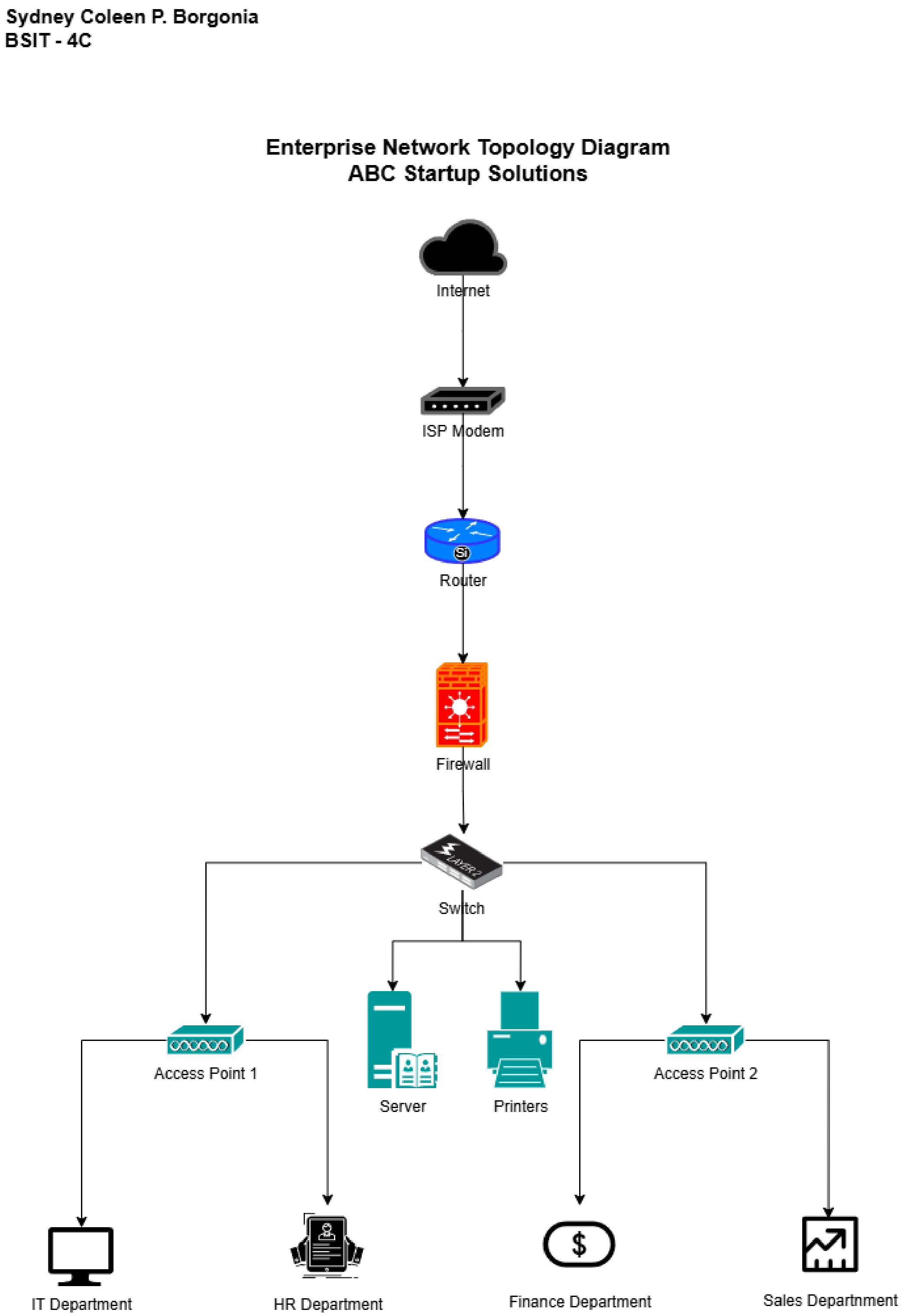
Asset ID	Network Equipment	Quantity	Purpose
NET-001	ISP Modem	1	Internet connection from ISP
NET-002	Router	1	Network routing and NAT
NET-003	Firewall	1	Network security and traffic filtering
NET-004	24-Port Managed Switch	1	Connect wired devices
NET-005	Wireless Access Point	3	Wireless connectivity
NET-006	24-Port Patch Panel	1	Structured cable management
NET-007	CAT6 Cable (305m box)	2	Network cabling
NET-008	RJ45 Connectors	50	Cable termination

Network Justification:

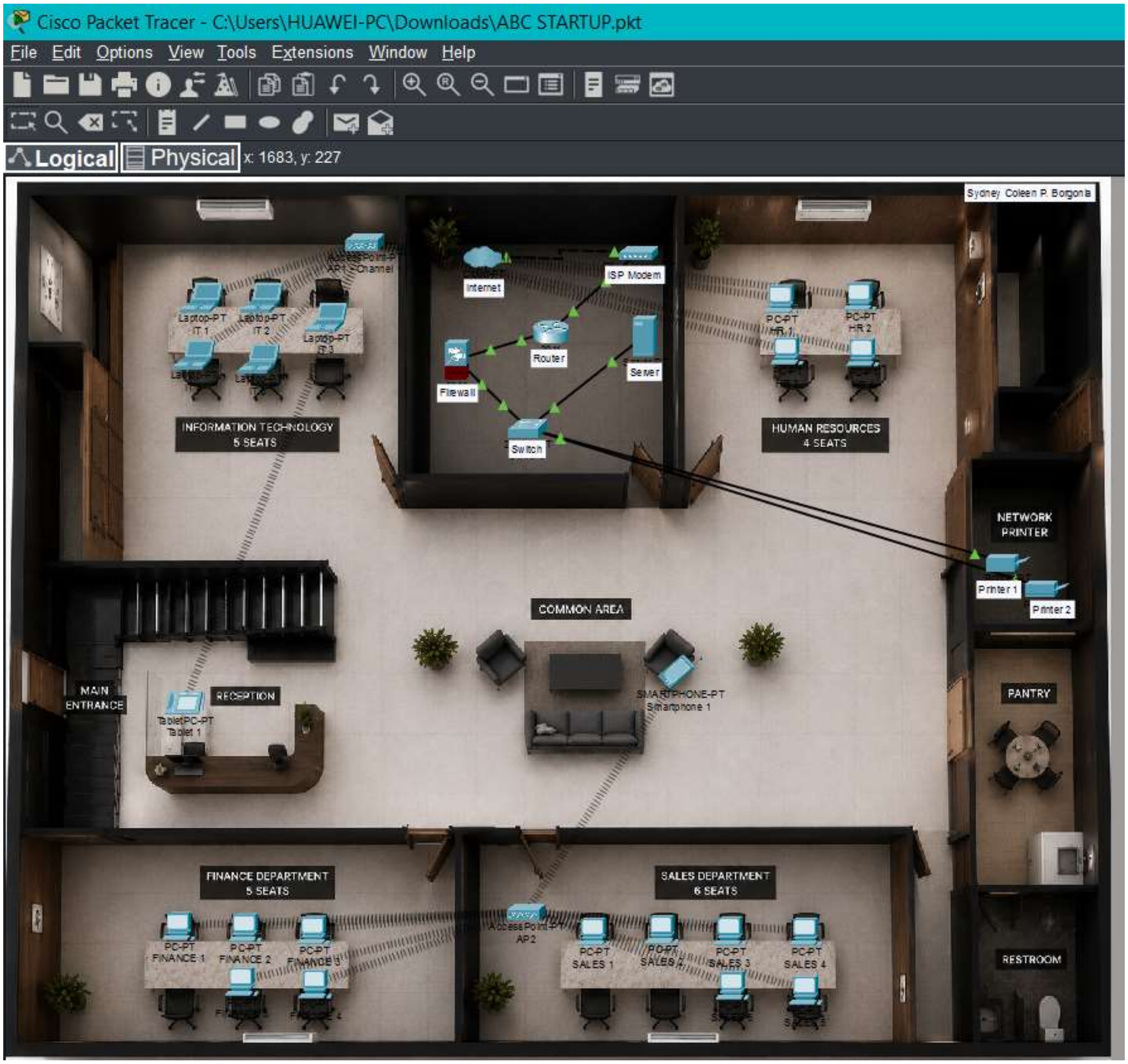
- **ISP Modem** – Provides the organization’s connection to the Internet and enables access to online services, cloud applications, and external communication.
- **Router** – Manages network traffic between the local network and the Internet while supporting NAT for secure and efficient Internet sharing among multiple devices.
- **Firewall** – Enhances network security by monitoring and filtering incoming and outgoing traffic to help prevent unauthorized access and cyber threats.
- **24-Port Managed Switch** – Connects computers, printers, servers, and other wired devices while supporting better network performance, monitoring, and future scalability.
- **Wireless Access Point** – Provides stable and secure wireless connectivity for employees using laptops, smartphones, and other mobile devices throughout the office.
- **24-Port Patch Panel** – Organizes and manages network cables properly, making maintenance, troubleshooting, and future upgrades easier.
- **CAT6 Cable** – Supports high-speed and reliable Ethernet communication between network devices and reduces signal interference within the office environment.

- **RJ45 Connectors** – Allows proper cable termination and ensures stable physical connections between switches, patch panels, and network devices.

Part 5: Enterprise Network Diagram



Cisco



Network Connection Description

The network starts with the Internet, which connects to the ISP Modem and then to the Router. The router directs network traffic to the Firewall (Cisco ASA 5505), which protects the internal network from unauthorized access.

The firewall connects to the Main Switch, which distributes the network connection to all departments and shared resources. The switch connects the 20 employee PCs, consisting of 5 for IT, 4 for HR, 5 for Finance, and 6 for Sales. It also connects the Server, Network Printer, and three Wireless Access Points.

Overall, the topology provides a centralized, secure, and organized network for all departments and network resources.

Connection Flow

Internet → ISP Modem → Router → Firewall → Main Switch → Department Devices

From the main switch, the network branches to:

- Information Technology – 5 Laptops
- Human Resources – 4 PCs
- Finance – 5 PCs
- Sales – 6 PCs
- Network Printer – 2
- Server – 1
- Wireless Access Points – 2

Part 6: System Administration Roles

Helpdesk Technician

Responsibilities

- Resolve user issues
- Install and configure software
- Provide technical support

Skills

- Troubleshooting
- Communication
- Basic networking

Common Tools

- AnyDesk
- Windows Tools
- Ticketing systems

Certifications

- CompTIA A+
- Google IT Support

Network Administrator

Responsibilities

- Manage routers and switches
- Configure network security

- Monitor network performance

Skills

- TCP/IP
- VLANs
- Firewall configuration

Common Tools

- Cisco Packet Tracer
- Wireshark
- Network monitoring tools

Certifications

- Cisco CCNA
- CompTIA Network+

Linux System Administrator

Responsibilities

- Manage Linux servers
- Configure services and backups
- Monitor server performance

Skills

- Linux command line
- Shell scripting
- Server management

Common Tools

- Ubuntu Server
- SSH
- Bash

Certifications

- LPIC-1
- Red Hat RHCSA

Cloud Administrator

Responsibilities

- Manage cloud resources
- Configure virtual servers and storage
- Implement backup and security policies

Skills

- Cloud platforms
- Virtualization

- Identity and access management

Common Tools

- AWS Management Console
- Microsoft Azure Portal
- Google Cloud Console

Certifications

- AWS Certified Cloud Practitioner
- Microsoft Azure Fundamentals (AZ-900)

Team Collaboration

These four professionals work together to maintain a reliable and secure IT environment. The Helpdesk Technician handles user support and escalates complex issues. The Network Administrator ensures stable connectivity and network security. The Linux System Administrator manages servers, services, and backups, while the Cloud Administrator oversees cloud resources and disaster recovery solutions. Their collaboration ensures that users can work efficiently, systems remain secure, and business operations continue without major interruptions.

Part 7: Infrastructure Recommendations

Internet Provider

- A fiber-optic business internet plan (300–500 Mbps) is recommended to support cloud services, video meetings, software development, and file transfers.

Server Specifications

- Intel Xeon or AMD EPYC processor
- 32 GB RAM
- 1 TB NVMe SSD (system)
- 4 TB RAID storage (data)
- Dual Gigabit Ethernet ports

Backup Strategy

- Daily automated backup to NAS
- Weekly backup to external drives
- Monthly cloud backup for critical company files

Security Recommendations

- Enable firewall protection
- Use Microsoft Defender on all endpoints
- Apply automatic operating system updates
- Implement role-based access control

Password Policy

- Minimum 12 characters
- Combination of uppercase, lowercase, numbers, and symbols

- Password expiration every 90 days
- Multi-factor authentication for administrative accounts

Expansion Plan

- The infrastructure should support growth to 50 employees by adding another managed switch, additional access points, increased server storage, and cloud-based backup services.

Part 8: Personal Reflection

This project helped me understand the importance of planning an IT infrastructure before purchasing equipment or deploying systems. I learned how hardware, software, networking, security, and documentation work together to support business operations. The most challenging task was designing the network topology because it required logical connections between devices and departments while maintaining security and scalability.

Planning is important because it prevents unnecessary expenses, reduces security risks, and ensures that future expansion can be implemented smoothly. By preparing inventories, diagrams, and infrastructure recommendations, I realized that a System Administrator must think beyond installing computers; they must consider reliability, performance, backup, and business continuity.

This project improved my skills in technical documentation, inventory management, network design, and infrastructure planning. It also strengthened my ability to present technical information in a professional and organized manner using GitHub and Markdown. In the future, these skills will help me become a more effective System Administrator who can design, maintain, and improve enterprise IT environments efficiently and securely.

References

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- Microsoft Defender Documentation: https://learn.microsoft.com/microsoft-365/security/defender-endpoint/